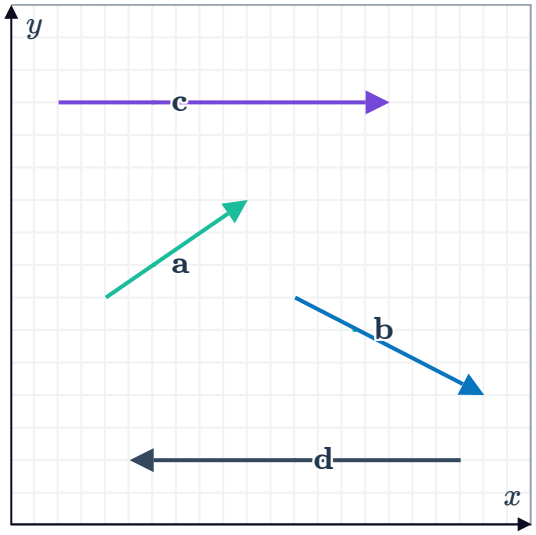
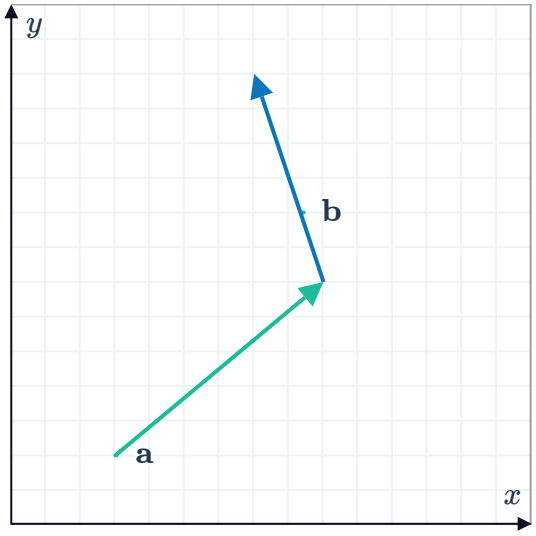
Geometric addition and subtraction of vectors (Extended)

1 Consider the following graph and identify which vector is the result of $\mathbf{a} + \mathbf{b}$:



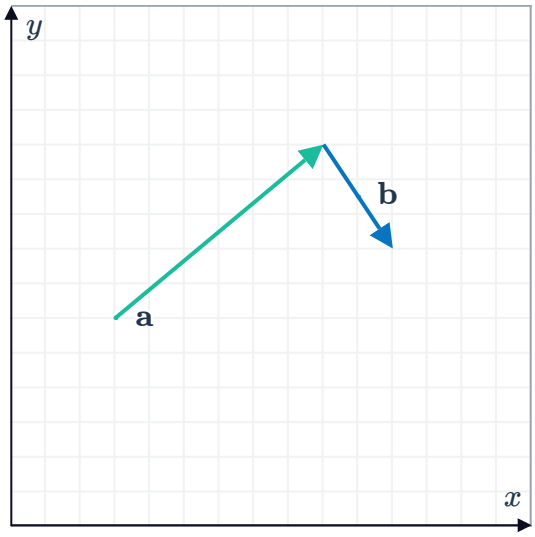
2 Consider the graph of vectors $\mathbf{a}$ and $\mathbf{b}$. Plot the result of $\mathbf{a} + \mathbf{b}$.



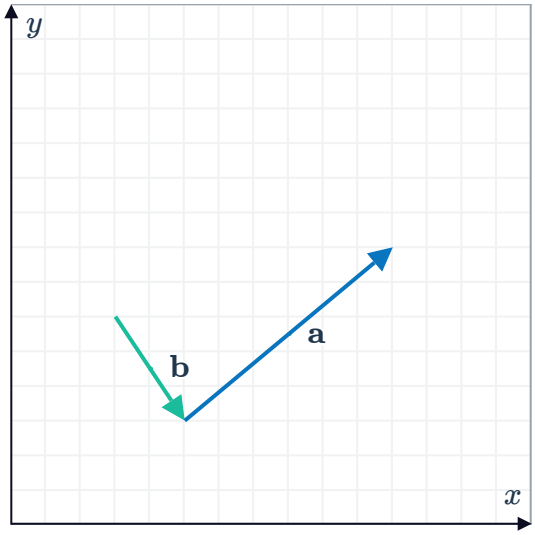
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3 Consider the following graphs of $\mathbf{a}$ and $\mathbf{b}$:

a Plot the result of $\mathbf{a} + \mathbf{b}$ in the first graph:



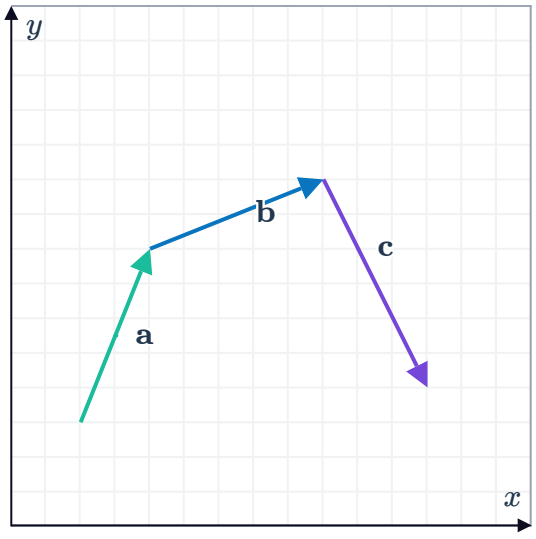
b Plot the result of $\mathbf{b} + \mathbf{a}$ in the second graph:



c Are the two resulting vectors $\mathbf{a} + \mathbf{b}$ and $\mathbf{b} + \mathbf{a}$ equal?

4 The vectors $\mathbf{a}$, $\mathbf{b}$ and $\mathbf{c}$ have been graphed as shown:

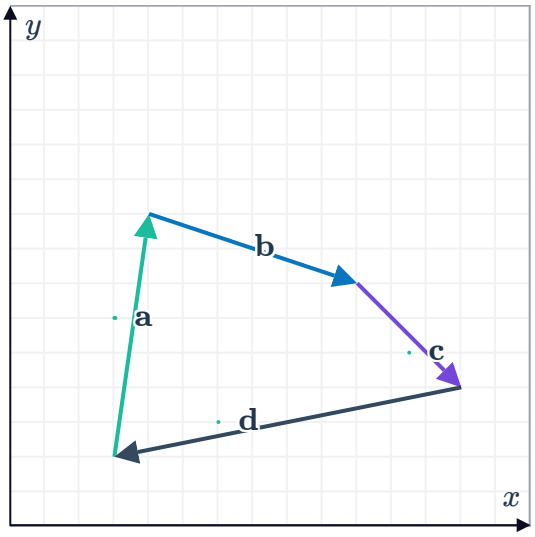
Plot the result of $\mathbf{a} + \mathbf{b} + \mathbf{c}$ on the same axes.



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5 Vectors $\mathbf{a}$, $\mathbf{b}$, $\mathbf{c}$ and $\mathbf{d}$ have been graphed as shown:

Plot the result of $\mathbf{b}+\mathbf{c}$ on the same axes.



Algebraic addition and subtraction of vectors

6 Rewrite $(6,5)+(7,-4)$ as a single vector.

7 For each of the following set of vectors, find:

i $\mathbf{A}+\mathbf{B}$ ii $\mathbf{B}+\mathbf{C}$

iii $\mathbf{A}+\mathbf{B}+\mathbf{C}$

a $\mathbf{A}=\begin{pmatrix}4\\9\end{pmatrix}, \mathbf{B}=\begin{pmatrix}8\\2\end{pmatrix}, \mathbf{C}=\begin{pmatrix}7\\5\end{pmatrix}$

b $\mathbf{A}=\begin{pmatrix}-6\\7\end{pmatrix}, \mathbf{B}=\begin{pmatrix}-4\\-8\end{pmatrix}, \mathbf{C}=\begin{pmatrix}2\\1\end{pmatrix}.$

c $\mathbf{A}=\begin{pmatrix}4.7\\-6.2\end{pmatrix}, \mathbf{B}=\begin{pmatrix}7.4\\2.3\end{pmatrix}, \mathbf{C}=\begin{pmatrix}-5.1\\-3.5\end{pmatrix}.$

d $\mathbf{A}=\begin{pmatrix}\frac{5}{6}\\\frac{6}{5}\end{pmatrix}, \mathbf{B}=\begin{pmatrix}-\frac{2}{5}\\-\frac{7}{5}\end{pmatrix}, \mathbf{C}=\begin{pmatrix}\frac{3}{2}\\-\frac{4}{5}\end{pmatrix}.$

Scalar multiplication of vectors

8 Let $\mathbf{a}=\begin{pmatrix}3\\1\end{pmatrix}$, find:

a $2\mathbf{a}$ b $5\mathbf{a}$ c $-1\mathbf{a}$ d $-6\mathbf{a}$

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9 Let $\mathbf{a} = \begin{pmatrix} 10 \\ 16 \end{pmatrix}$, find:

a $3\mathbf{a}$ b $-10\mathbf{a}$ c $\frac{1}{2}\mathbf{a}$ d $-\frac{1}{4}\mathbf{a}$

10 Let $\mathbf{a} = \begin{pmatrix} -4 \\ 9 \end{pmatrix}$, find:

a $3\mathbf{a}$ b $-2\mathbf{a}$ c $\frac{1}{2}\mathbf{a}$ d $-100\mathbf{a}$

Scalar multiplication of vectors (Extended)

11 Let $\mathbf{a} = \begin{pmatrix} 2 \\ 4 \end{pmatrix}$. Using the origin as the starting point, plot the following vectors:

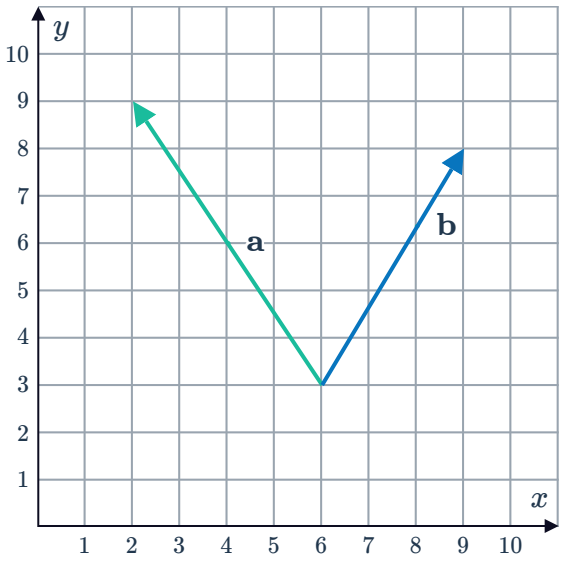
a $3\mathbf{a}$ b $-2\mathbf{a}$ c $\frac{1}{2}\mathbf{a}$

12 Consider vectors $\mathbf{a}$ and $\mathbf{b}$ plotted on the graph below. Find the column vector for each of the following:

a $\mathbf{a} + \mathbf{b}$ b $3\mathbf{b}$

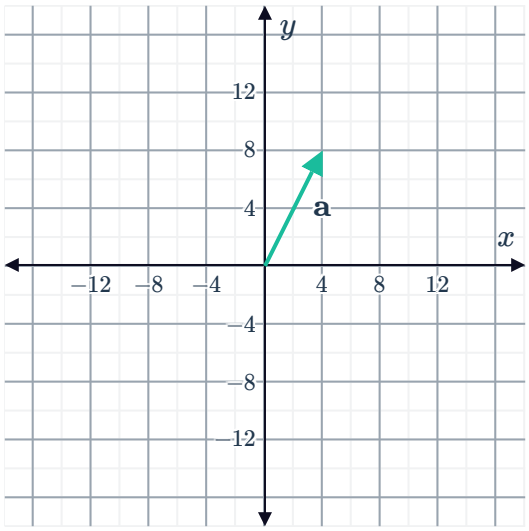
c $4\mathbf{a}$ d $5\mathbf{b} - \mathbf{a}$

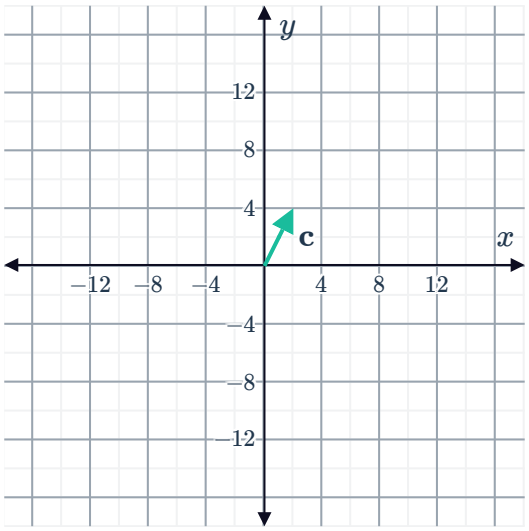
e $3\mathbf{a} - 4\mathbf{b}$

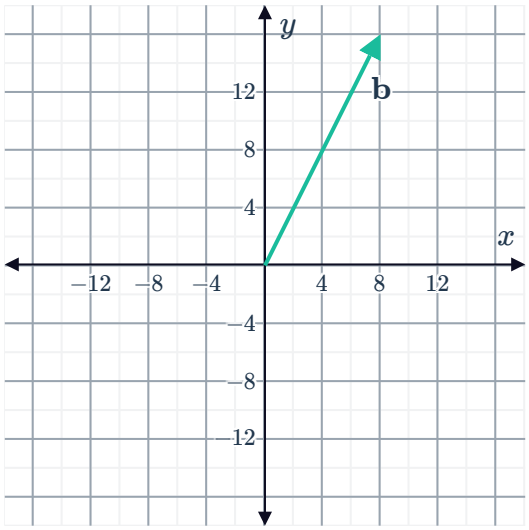


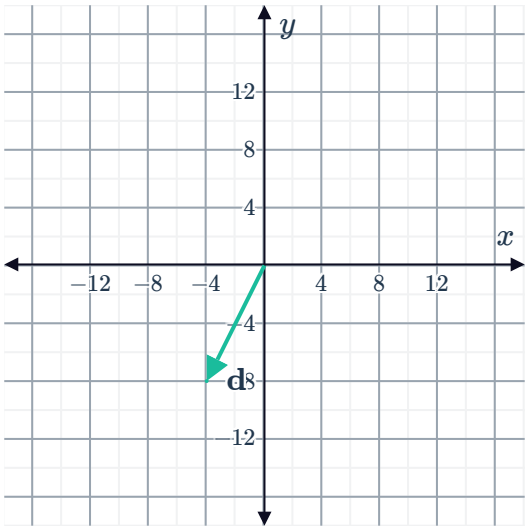
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13 Consider the vectors shown below:









a Which vector is equivalent to $2\mathbf{a}$?

b Which vector is equivalent to $\frac{1}{4}\mathbf{b}$?

c Which vector is equivalent to $-\mathbf{a}$?

d Which vector is equivalent to $-\frac{1}{2}\mathbf{d}$?

14 Consider the vectors plotted on the graph. Find the column vector for each of the following:

a $3\mathbf{a}$

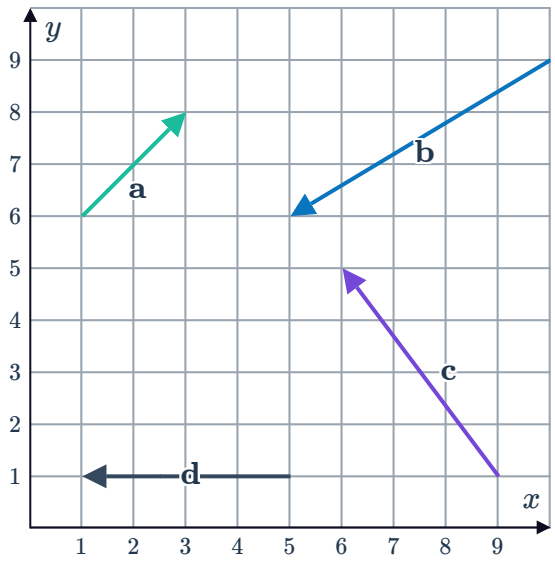
b $4\mathbf{c}$

c $2\mathbf{b} + \mathbf{d}$

d $\mathbf{c} - \mathbf{d}$

e $3\mathbf{a} - \mathbf{b}$

f $-4\mathbf{b} + 2\mathbf{c}$



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15 If $\mathbf{a} = (2x, 8x)$, $\mathbf{b} = (3x, 4x)$, and $|\mathbf{a} + \mathbf{b}| = 13$, find the value of x.

16 If $\mathbf{a} = (2x, 4)$, $\mathbf{b} = (3x, 8x)$, and $|\mathbf{a} + \mathbf{b}| = 13x$, find the value of x.

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